

CS31: Graphs

Main reference : *Discrete mathematics and Its Applications*, 7th edition, Kenneth H. Rosen

Other reference : *Discrete mathematics with Applications*, 4th edition, Susanna S. Epp

Other reference : *Discrete mathematics: An Introduction to Proofs and Combinatorics*, Kevin Ferland

1 Introduction

Once you know the problem, you are half the solution.

This is an old saying but this is a good motivation for why one should be interested in understanding the concept of graphs. When solving complex problems, we first understand the problem. Part of understanding is creating a simplistic model of the problem so that other information are initially masked and only the important details are retained. One relatively simple model called *graph* is mainly composed of: (a) dots and (b) lines connecting the dots. Graphs are widely-used and have a lot of real-life applications as in the following.

Imagine an organization that wants to set up teams of three to work on some projects. In order to maximize the number of people on each team who had previous experience working together successfully, the director asked the members to provide names of their past partners. This information is displayed in the table given in Figure 1. Given the graph in the figure, each dot represents a person and a connection from one person to another indicates that the connected dots correspond to past partners. The goal then is to find groups of three connected dots.

Name	PastPartners
Ana	Dan, Flo
Bev	Cai, Flo, Hal
Cai	Bev, Flo
Dan	Ana, Ed
Ed	Dan, Hal
Flo	Cai, Bev, Ana
Gia	Hal
Hal	Gia, Ed, Bev, Ira
Ira	Hal

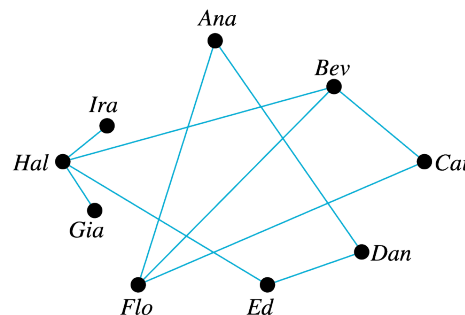


Figure 1: An example of a graph representation (from (Epp, 2011))

Telephone, electric power, gas pipeline, and air transport systems can all be represented by graphs, as can computer networks – from small local area networks to the global Internet system that connects millions of computers worldwide. Questions that arise in the design of such systems involve choosing connecting edges to minimize cost, optimize a certain type of service, and so forth. A typical network, called a hub and spoke model, is shown below.

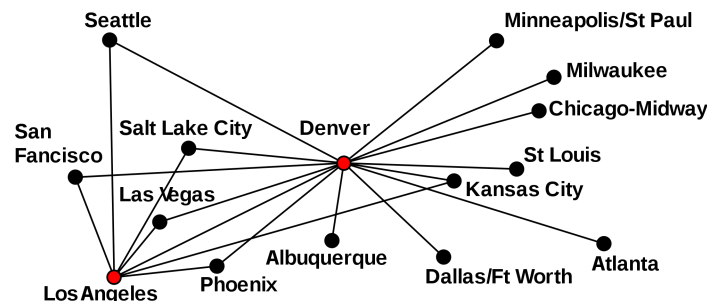


Figure 2: An example hub and spoke model (from commons.wikimedia.org)

Another popular network is the World Wide Web, a system of interlinked documents, or webpages, contained on the Internet. Users employing Web browsers, such as Internet Explorer, Google Chrome, Apple Safari, and Opera, can move quickly from one webpage to another by clicking on hyperlinks, which use versions of software called hypertext transfer protocols (HTTPs). Individuals and individual companies create the pages, which they transmit to servers that contain software capable of delivering them to those who request them through a Web browser. Because the amount of information currently on the Web is so vast, search engines, such as Google, Yahoo, and Bing, have algorithms for finding information very efficiently.

The idea of a graph started from a puzzle in an old town named Königsberg, Prussia (now Kaliningrad, Russia). As shown in the left of Figure 3, the puzzle involves taking a tour in the beautiful town of Königsberg. Starting in any of the district of the city of Königsberg, is it possible to go for a walk that passes over every bridge exactly once, and return to the starting district? The puzzle is also known as the Königsberg bridge problem. A graph representation of the seven bridges is shown in the right side of Figure 3. The short answer to the question above is no. However, realizing this answer was not easy. Through the efforts of a mathematician by the name of Leonhard Euler, his results on this problem paved the way for graph theory. (We will revisit this problem at a later topic)

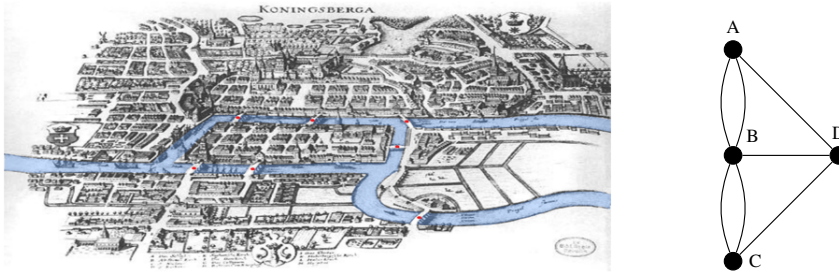


Figure 3: Left figure gives an illustration of the town of Königsberg (adapted from wikipedia.org). The right figure is a graph representation of the 7 bridges of Königsberg.

2 Graph Terminologies

We first provide basic terminologies regarding graphs. We will be using these terminologies as we explore different problems and properties of graphs.

2.1 Basic terminologies

We first provide the following definition of a graph. Note here that the dots are called *vertices* and the connections between dots are called *edges*.

Definition 1 A (*undirected*) *graph*, G , is defined as a pair:

$$G = (V, E),$$

where:

- V is a finite non-empty set of vertices.
- E is a finite set of edges.
- The edge-endpoint function to G , denoted as ϵ , relates each edge to a corresponding set of one or two vertices. Let $\mathbb{P}_k(V)$ be the set of subsets of V where each subset has exactly k elements. Then, $\epsilon : E \rightarrow (\mathbb{P}_1(V) \cup \mathbb{P}_2(V))$

- For some $e \in E$, and $u, v \in V$, if $e \mapsto \{u, v\}$, then vertices u and v are said to be the **endpoints** of edge e , and vertices u and v are said to be **adjacent**. An edge is said to be **incident** with each of its endpoint. Thus, edge e is incident both with u and v .
- If $e \mapsto \{u\}$, that is edge e only has one endpoint, u , then e is said to be a **loop**. The vertex u is said to be **adjacent to itself**.
- If $e_1 \mapsto \{u, v\}$ and $e_2 \mapsto \{u, v\}$, that is e_1 and e_2 have the same endpoints, then e_1 and e_2 are said to be **parallel**.

The **order** of graph G is the number of its vertices while the **size** of graph G is the number of edges.

Note: A graph is said to be trivial if it contains only one vertex and no edge.

Example 1 Let us label the graph in Figure 4 as graph G . Then, $G = \{V, E\}$ where

- $V = \{v_1, v_2, v_3, v_4\}$
- $E = \{e_1, e_2, e_3, e_4, e_5\}$
- $\epsilon = \{e_1 \mapsto \{v_1, v_4\}, e_2 \mapsto \{v_2, v_3\}, e_3 \mapsto \{v_2, v_3\}, e_4 \mapsto \{v_3, v_4\}, e_5 \mapsto \{v_4\}\}$

Thus, the order of graph G is four while its size is five. Applying the concept of endpoints, since e_4 connects v_3 and v_4 , i.e. $e_4 \mapsto \{v_3, v_4\}$, we refer to v_3 and v_4 as endpoints of edge e_4 . We then say, e_4 is incident with both v_3 and v_4 . We also say that v_3 is adjacent to v_4 and vice versa. Graph G contains a loop, namely e_5 . As can be observed, the edge e_5 points to and from the same vertex, i.e. v_4 . Graph G also contains parallel edges, namely e_2 and e_3 . Notice that both edges connect vertices v_2 and v_3 .

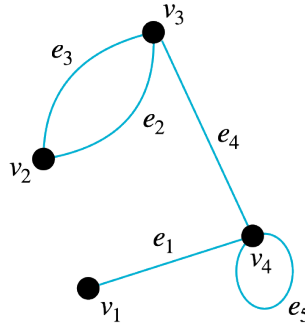


Figure 4: An example graph with four vertices and five edges (from (Epp, 2011))

A graph is usually visually represented via a drawing:

Definition 2 A **drawing** of a graph $G = (V, E)$ in the plane is a one-to-one assignment of the vertices to points in the plane and, for each edge, the assignment of a curve joining the ends of the edge in such a way that

1. the only vertex points hit by a curve are the endpoints of the edge it represents,
2. each curve is one-to-one (a curve does not intersect with itself) with the exception that the ends of a loop are assigned a common point, and
3. the images of curves associated with two distinct edges intersect at most finitely many points.

An intersection of two curves outside their endpoints is called a **crossing**.

Note that there can be many possible drawings that refer to the same graph. What is important is the connections made by the edges.

Example 2 Give a drawing for the following graph definition: Let $G = (V, E)$ be a graph.

- $V = \{v_1, v_2, v_3, v_4, v_5\}$
- $E = \{e_1, e_2, e_3, e_4, e_5\}$
- $\epsilon = \{e_1 \mapsto \{v_1, v_2\}, e_2 \mapsto \{v_2, v_3\}, e_3 \mapsto \{v_3, v_4\}, e_4 \mapsto \{v_4, v_5\}, e_5 \mapsto \{v_5, v_1\}\}$

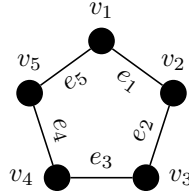


Figure 5: A drawing for the graph described in Example 2.

In some cases, it may be important to assign a direction to an edge. For example, in modeling the data flow in a computer network, one may need to distinguish the flow of data from a point A to a point B and vice versa. To represent such, we use a directed graph where an edge is associated to an ordered pair of vertices.

Definition 3 A **directed graph**, G , is defined as:

$$G = (V, D)$$

where:

- V is a finite non-empty set of vertices.
- D is a finite set of directed edges. The directed edge-endpoint function $d\epsilon : D \rightarrow V \times V$, associates each directed edge to an ordered pair of vertices.
- For some $d \in D$, and $u, v \in V$, if $d \mapsto (u, v) \in d\epsilon$, then d is a directed edge from vertex u to vertex v . We refer to u as the **initial** vertex of d and v is called the **terminal** vertex of d .

In drawing directed graphs, we represent a directed edge as an arrow.

Example 3 Give a drawing for the graph $G = (V, D)$ defined as:

- $V = \{v_1, v_2, v_3, v_4, v_5\}$,
- $D = \{d_1, d_2, d_3, d_4, d_5, d_6, d_7, d_8\}$,
- $d\epsilon = \{d_1 \mapsto (v_1, v_2), d_2 \mapsto (v_2, v_1), d_3 \mapsto (v_1, v_5), d_4 \mapsto (v_5, v_1), d_5 \mapsto (v_5, v_4), d_6 \mapsto (v_4, v_5), d_7 \mapsto (v_3, v_4), d_8 \mapsto (v_4, v_3)\}$

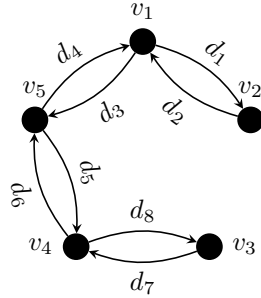


Figure 6: A drawing for the graph described in Example 3.

2.2 Neighborhood

Let's now introduce terminologies pertaining to vertices adjacent to a particular vertex of a graph.

Definition 4 The set of all vertices adjacent to a vertex v of a graph $G = (V, E)$, denoted by $N(v)$, is called the **neighborhood of v** . If A is a subset of V , we denote by $N(A)$ the set of all vertices in G that are adjacent to at least one vertex in A , i.e. $N(A) = \bigcup_{v \in A} N(v)$.

Definition 5 (Degrees in Graph) Let $G = (V, E)$ be a graph.

- The **degree** of a vertex v in G , denoted as $\deg(v)$, is the number of edges incident with v . Loops are counted twice.
- The **maximum degree** of G is the degree of the vertex in G with the most number of incident edges.
- The **total degree** of G is the sum of the degrees of all vertices in G .

Example 4 Let's consider the example graph in Figure 4. What are the degrees and neighborhoods of the vertices in the graph?

Answer: The degree of each vertex is as follows: $\deg(v_1) = 1$, $\deg(v_2) = 2$, $\deg(v_3) = 3$, $\deg(v_4) = 4$. Note that the loop in v_4 is counted twice. Thus, the maximum degree of the graph is four, and the total degree is $1 + 2 + 3 + 4 = 10$.

The neighborhood of each vertex is as follows: $N(v_1) = \{v_4\}$, $N(v_2) = \{v_3\}$, $N(v_3) = \{v_2, v_4\}$, $N(v_4) = \{v_1, v_3, v_4\}$. Note that $v_4 \in N(v_4)$. Thus, observe that if v is in $N(v)$, then a loop occurs at vertex v .

Imagine that a graph is a party where each vertex represents a person and an edge represents two people shaking hands. We can then realize that when counting the sum of the degrees of all vertices in a graph, each edge contributes twice. This property is given in the following theorem.

Theorem 1 The Handshake Theorem

Given an undirected graph $G = (V, E)$

$$\sum_{u \in V} \deg(u) = 2|E|$$

This theorem provides us a way to compute edges given the sum of the degrees of all vertices. How many edges are there in a graph with three vertices of degree four? The handshaking theorem tells us that $2 \mid \sum \deg(u) = 3(4)$. Therefore, $\sum \deg(u) = 6$. How many edges are there in a graph with five vertices of degree three? The handshaking theorem tells us that this is not possible because the sum of the degrees will then be $5(3) = 15$ which is an odd number.

Corollary 1 *Every undirected graph has an even number of vertices with odd degree.*

Proof: Consider V_1 as the number of vertices with even degree and V_2 as the number of vertices with odd degree. Then, from Theorem 1,

$$\sum_{u \in V_1} \deg(u) + \sum_{u \in V_2} \deg(u) = 2|E|$$

Note that even numbers are closed under addition, thus, $\sum_{u \in V_1} \deg(u)$ is even. In order to obtain an even number in the right-hand side of the above equation, $\sum_{u \in V_2} \deg(u)$ must also be even. Since V_1 is composed of vertices with odd degree, we know that we are adding odd numbers in $\sum_{u \in V_2} \deg(u)$. In order for a summation of odd numbers to be even, the total of odd numbers being summed must also be even. ■

Test Your Knowledge

1. Does an undirected graph with four vertices of degree 1, 1, 2, and 3, exist?
2. Does an undirected graph with four vertices of degree 1, 1, 3, and 3, exist?

In the context of directed graphs, we distinguish the number of incoming edges and outgoing edges of a vertex.

Definition 6 (Degrees in Directed Graph) Let $G = (V, D)$ be a directed graph. For vertices $u, v \in V$,

- The in-degree of a vertex v , denoted by $\text{indeg}(v)$, represents the number of all incoming incident edges of v .
- The out-degree of a vertex v , denoted by $\text{outdeg}(v)$, represents the number of all outgoing incident edges of v .
- A loop contributes to one incoming edge, and one outgoing edge.

Example 5 Consider the directed graph in Example 3. What are the in-degrees and out-degrees of the vertices in the graph?

Answer: The in-degree of each vertex is as follows: $\text{indeg}(v_1) = \text{indeg}(v_4) = \text{indeg}(v_5) = 2$, $\text{indeg}(v_2) = \text{indeg}(v_3) = 1$. The out-degree of each vertex is the same as its in-degree, that is, $\text{outdeg}(v_1) = \text{outdeg}(v_4) = \text{outdeg}(v_5) = 2$, $\text{outdeg}(v_2) = \text{outdeg}(v_3) = 1$.

Observe that the total in-degree of all vertices is the same as the total out-degree of all vertices, that is, $2 + 2 + 2 + 1 + 1 = 8$. This value is equal to the number of edges.

Every edge has an initial vertex and a terminal vertex. Thus, the sum of the in-degrees of all vertices and the sum of the out-degree of all vertices are the same. This value is equal to the number of edges in the graph.

Theorem 2 For any directed graph $G = (V, D)$,

$$\sum_{v \in V} \text{indeg}(v) = \sum_{v \in V} \text{outdeg}(v) = |D|$$

2.3 Special Types of Graphs

We now study terms used to distinguish graphs with different features. We start with some graph classification depending on the presence of parallel edges and loops.

Definition 7

- **Pseudographs** - Graphs that allow parallel edges and loops.
- **Multigraphs** - Graphs that may have parallel edges.
- **Simple Graphs** - Graphs that do not have loops nor parallel edges. An undirected edge with endpoints v and w is then denoted by $\{v, w\}$ while a directed edge from v to w is denoted by (v, w) .

Simple graphs are one of the most common and well-studied type of graph. In such graphs, an edge is simply denoted by the pair of vertices connected to it.

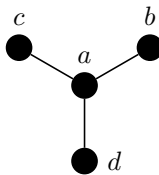
Definition 8 Given below are the definitions of other special kinds of simple graphs based on construction:

- **Complete Graph** - Let $n \in \mathbb{Z}^+$. A complete graph of n vertices, denoted as K_n , is a simple graph with n vertices where there is exactly one edge connecting every pair of distinct vertices.
- **Path Graph** - A path graph of n vertices, denoted as P_n , is a simple graph $G = (V, E)$, where:
 - $V = \{v_1, \dots, v_n\}$, and
 - $E = \{\{v_i, v_{i+1}\} \mid \forall i \in \{1, \dots, n-1\}\}$
- **Cycle Graph** - A cycle graph of n vertices, denoted as C_n , is a simple graph $G = (V, E)$, where:
 - $V = \{v_1, \dots, v_n\}$, and
 - $E = \{\{v_i, v_{i+1}\} \mid \forall i \in \{1, \dots, n-1\}\} \cup \{\{v_1, v_n\}\}$
- A **regular graph** is a simple graph where all its vertices have the same degree. A **k -regular graph** is a regular graph where all its vertices have degree k .

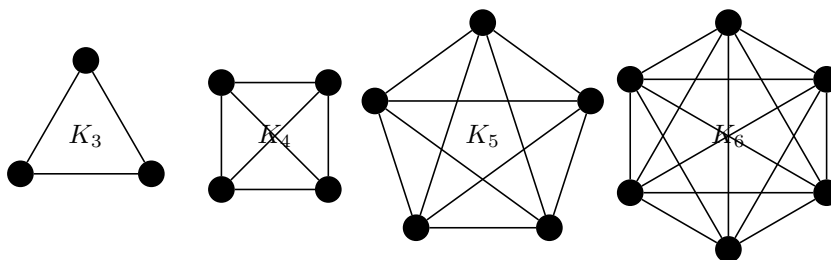
Example 6

1. Give a drawing for a simple graph $G = (V, E)$

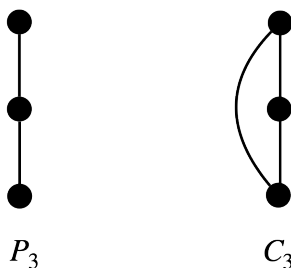
- $V = \{a, b, c, d\}$, and
- $E = \{\{a, b\}, \{a, c\}, \{a, d\}\}$.



2. Give a drawing for complete graphs K_3 , K_4 , K_5 , and K_6 .



3. Give a drawing for a path graph P_3 and a cycle graph C_3



Notice that although there are different drawings for K_3 (in the previous item) and C_3 . Both drawings refer to the same graph.

4. The graphs in the first and second item are all examples of a regular graph. As can be observed, a complete graph is always a regular graph.

Another interesting special type of a simple graph is a bipartite graph where the vertex set can be partitioned into two disjoint subsets and edges are only allowed to connect two vertices belonging to different subsets.

Definition 9

- **Bipartite graph** is a simple graph $G = (V_1 \cup V_2, E)$ such that $V_1 \cap V_2 = \emptyset$, and the edges of E are of the form $\{u, v\}$, where $u \in V_1$, and $v \in V_2$. There is no edge in E that has endpoints belonging to the same vertex subsets.
- A **complete bipartite graph** is a graph of $m \times n$ vertices, denoted as $K_{m,n}$ is a bipartite graph where each vertex in V_1 is connected to all vertices in V_2 .

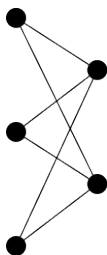


Figure 7: A $K_{3,2}$ complete bipartite graph.

Bipartite graphs can be used to model matching problems. Consider the case where each of the two disjoint subsets represent job hunters and prospect companies and an edge is essentially a possible hiring. Another example is modelling job assignments, that is, the two subsets can represent the jobs and workers and each edge represents a possible assignment.

Test Your Knowledge

- How many edges are there in P_n ? C_n ? K_n ? For K_n , there are $\binom{n}{2}$.
- Does a simple undirected graph with four vertices of degree 1, 1, 3, and 3 exist?
- Is C_6 a bipartite?
- Is C_3 a bipartite?

In some applications, the graph required needs a numerical value to characterize an edge in a graph. This is relevant for example when modeling road networks just like in taxi-hailing apps. We can imagine a graph where each vertex is a location and an edge is essentially a travel from one location to another. When determining how to get to a destination in a short amount of time, we need to indicate how long it takes to travel from one location to another. This leads us to the following definition.

Definition 10 A **weighted graph** is a graph where its edges are mapped to real numbers by a function called the cost function so that $c : E \rightarrow \mathbb{R}$.

Figure 8 provides a drawing of an example weighted graph. The numbers in blue show the weights associated with the individual edges.

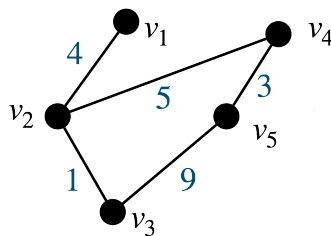


Figure 8: An example of a weighted graph

2.4 Subgraphs

When solving problems involving graphs, it may happen that a certain approach doesn't need the whole graph but only a part of the graph. For example when modeling the network of friends in a social networking site and tracking the sphere of influence of a group of people, we may want to consider only the part of the graph that is relevant to the group.

From an original graph, when we delete some vertices and edges without removing endpoints of any remaining edges, we still obtain a graph, albeit a smaller one. We thus call this a subgraph.

Definition 11 A **subgraph of a graph** $G = (V, E)$ is a graph $H = (W, F)$ where $W \subseteq V$ and $F \subseteq E$. A subgraph H of G is a proper subgraph of G if $H \neq G$.

We also define a subgraph that can be formed by a subset of the vertex set as follows:

Definition 12 Let $G = (V, E)$ be a simple graph. The **subgraph induced by** a subset W of the vertex set V is the graph (W, F) , where the edge set F contains an edge in E if and only if both endpoints of this edge are in W .

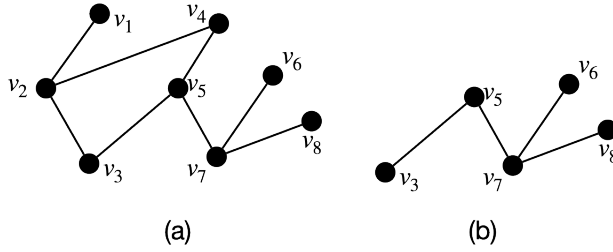


Figure 9: Graph in (b) is a subgraph induced by subset $W = \{v_3, v_5, v_6, v_7, v_8\}$ of the vertex set in (a)

Multiple small graphs can also be joined to form a single large graph as given in the ff. definition:

Definition 13 The **union** of two simple graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ is the simple graph with vertex set $V_1 \cup V_2$ and edge set $E_1 \cup E_2$. The union of G_1 and G_2 is denoted by $G_1 \cup G_2$.

3 Graph Tours

One of the more interesting questions on graphs involves travelling in a graph and finding a route from one vertex to another. Recall that this is actually the topic concerning the Königsberg problem whose results by the great mathematician Leonhard Euler paved the way for graph theory.

Definition 14 (Graph Tours)

Let n be a nonnegative integer and let $G = (V, E)$ be a graph. A **path of length n from vertex u to vertex v in G** is a sequence of n edges

$$e_1, \dots, e_n$$

of G for which there exists a sequence

$$x_0 = u, x_1, \dots, x_{n-1}, x_n = v$$

of vertices such that e_i has x_{i-1} and x_i for $i = 1, \dots, n$.

- When the graph is simple, we denote this path by its vertex sequence $x_0, x_1, \dots, x_n$ (since listing the vertices uniquely determines the path).
- The path is a **circuit** if it begins and ends at the same vertex ($u = v$) and has length greater than zero.
- The path or circuit is said to **pass through** the vertices $x_1, x_2, \dots, x_{n-1}$ and **traverse** the edges $e_1, \dots, e_n$.
- A path or circuit is **simple** if it does not contain the same edge more than once.
- The **distance** between two vertices, u and v , in G , denoted as $\text{dist}(u, v)$, is the length of the shortest path in G between u and v . If there is no path from u to v , then $\text{dist}_G(u, v) = \infty$.

Remark 1: Notice that this terminology is also readily extended to directed graphs.

Remark 2: The terms used in graph theory have not yet been standardized. Hence, the terms used in this definition might be named differently in other references. Be sure to check the definitions in the resource you are using.

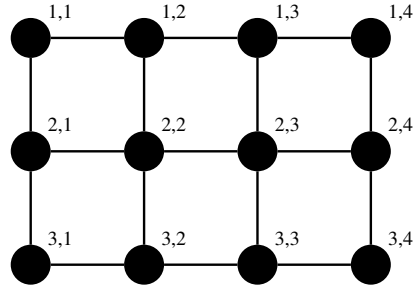


Figure 10: Graph G_M , a 3 row, 4 column vertex grid.

Example 7 Consider the drawing of graph G_M :

Note that since graph G_M is a simple graph, we can specify a path as a sequence of vertices (instead of a sequence of edges).

1. Give a **path** from vertex $(1,1)$ to vertex $(3,4)$ in G_M .
One possible answer is $(1,1), (2,1), (2,2), (2,3), (2,4), (3,4)$. Note that this is also an example of a simple path.
The path $(1,1), (2,1), (2,2), (2,3), (2,4), (1,4), (1,3), (2,3), (2,4), (3,4)$ is another possible answer. Since the vertices $(2,3)$ and $(2,4)$ are repeated, this answer is not a simple path.
2. Give a **circuit** containing vertex $(2,1)$ with length 8 in G_M .
One possible answer is $(2,1), (2,2), (2,3), (3,3), (3,2), (3,1), (2,1)$.
3. What is the **distance** between vertex $(1,1)$ and vertex $(3,4)$ in G_M ?
The answer is 5. Notice that we need three at the minimum, we need three edges from column 1 to column 4. We also need two edges to reach from row 1 to row 3. One possible path with length 5 from $(1,1)$ to $(3,4)$ is $(1,1), (2,1), (3,1), (3,2), (3,3), (3,4)$.

Definition 15

Let $G = (V, E)$ be a graph.

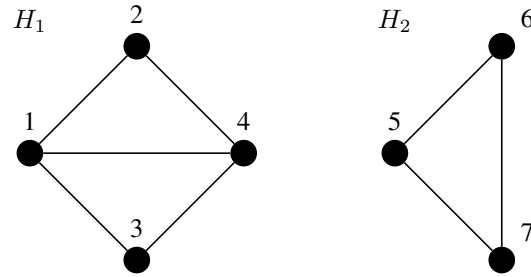
- Two vertices u and v of G are **connected** if, and only if, there is a path from u to v .
- G is said to be **connected** if, and only if, each pair of vertices in G is connected. Otherwise, G is **disconnected**.
- A **component** of a graph G is a connected subgraph H that is not contained in any other connected subgraph of G . Two vertices are in the same component if and only if there is a path between them.

Given below are some facts relating circuits and connectedness. Let $G = (V, E)$ be a graph:

- If G is connected, then any two distinct vertices of G can be connected by a simple path.
- If $u, v \in V$, and u and v belong to a simple circuit in G and one edge is removed from the cycle, then, there is still a path from u to v in G .
- If G is connected and contains a simple circuit, then an edge of the circuit can be removed without disconnecting G .

Example 8

Consider the drawing of graph G_C in Figure 11. Graph G_C is a disconnected graph. How

Figure 11: Graph G_C

many components make up graph G_C ?

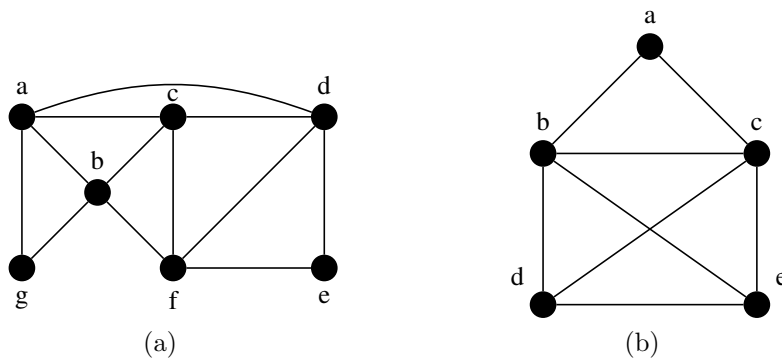
The answer is two. One component is represented by the subgraph H_1 . The other component is represented by the subgraph H_2 . Note that all vertices in H_1 is reachable from all other vertices in H_1 . Same goes for H_2 .

Definition 16 (Euler circuit and path)

Let G be a graph.

- An **Euler circuit** in graph G is a simple circuit containing every edge of G .
- An **Euler path** in graph G is a simple path containing every edge of G .

Example 9 Consider the graph drawings of G_{EC} and G_{EP} in Figure 12. Provide an example Euler circuit for graph G_{EC} and an example Euler path for graph G_{EP} .

Figure 12: Drawing in (a) represents graph G_{EC} while drawing in (b) represents graph G_{EP}

Verify that the following is an Euler circuit in graph G_{EP} : $a, b, c, d, e, f, d, a, c, f, b, g, a$. Also, Verify that the following is an Euler path in graph G_{EP} : $d, e, c, a, b, c, d, b, e$.

Test Your Understanding

Refer to Example 9. Is there an Eulerian path in G_{EC} ? How about an Eulerian circuit in graph G_{EP} ?

Let us observe the Euler circuit we identified earlier in Example 9. We are particularly interested in the degree of vertices in a graph with an Euler circuit.

The circuit starts with a vertex a and continues with an edge incident with a , i.e. $\{a, b\}$. Thus, edge $\{a, b\}$ contributed one to the degree of vertex a . Everytime the circuit passes

through a vertex, it contributed two to the vertex's degree. For example, the first time the circuit passes through vertex b , the circuit traverses edge $\{a, b\}$ and then edge $\{b, c\}$. The second time the circuit passes through vertex b , the circuit traverses edge $\{f, b\}$ and then eventually leaves vertex b by traversing $\{b, g\}$. This is true in all vertices passed through while the circuit has not yet terminated. Finally, the circuit terminates by traversing an edge incident to vertex a , i.e. $\{g, a\}$. This edge then contributes one to the degree of vertex a . Thus, the degree of vertex a is even. Also, all other vertices have an even degree.

This behavior holds for any Euler circuit. Thus, we have the following theorem:

Theorem 3 *If a graph G has an Euler circuit, then every vertex of the graph has positive even degree.*

Corollary 2 *If a graph G has a vertex with odd degree, then G has no Euler circuit.*

Question: If all the vertices of a graph G has positive even degree, is G Eulerian? We know that this is not true if graph G is not connected. How about if graph G is connected? To provide an affirmative response, we need to provide an algorithm that can be used to obtain an Euler circuit in a graph G where all vertices have positive even degrees.

We can start with any vertex a of G at which to start. We first construct an initial simple circuit that starts and ends with vertex a . Let the resulting circuit be denoted by C . What do we know about C ? Since each vertex has an even degree, there must be an even number of edges incident with this vertex in C . If all edges have been used up, an Euler circuit has already been constructed. Otherwise, we consider the subgraph H obtained from G by deleting the edges already used.

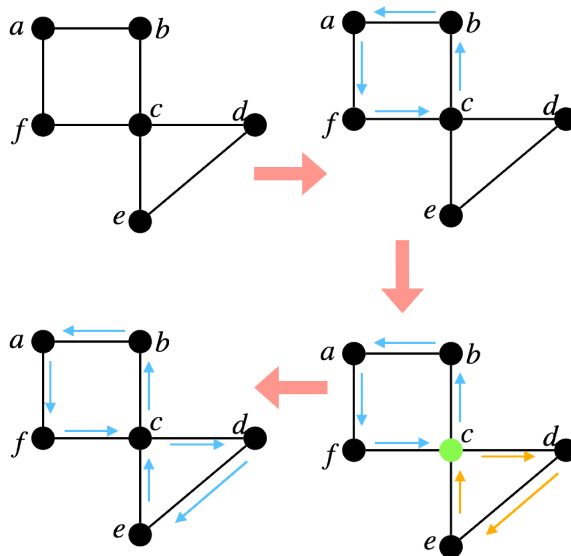


Figure 13: An illustration of how we can obtain a Euler circuit from a connected graph with even-degree vertices.

What do we know about H ? Since G is connected, H must have at least one vertex in common with the circuit that has been deleted. Also, every vertex in H must have even degree because all the vertices in G have even degree and for each vertex appearing in C , a pair of edges incident with this vertex have been removed. We then create another simple circuit, say C' , from H that starts and ends with a shared vertex. We then join C and C' by inserting C' at the appropriate vertex in C to form a new circuit. If all edges have already been used up, we have already produced an Euler circuit, otherwise, we again construct a new H and C' and continuously do so until all edges have been used up.

Theorem 6 If C is a Hamilton circuit in a graph $G = (V, E)$, then

C covers exactly two edges incident with each vertex, and

C has no subgraph which is a simple circuit on fewer than $|V|$ vertices.

From Theorem 4, we can easily determine whether a given graph has an Euler circuit. Unfortunately, this is not the case for Hamilton circuit. Specifically, there is no known efficient algorithm to determine if there is a Hamilton circuit in a given graph.

However, there are simple techniques we can use to determine if a graph does not have a Hamilton circuit. For example, we can profile the subgraph obtained from the resulting Hamilton circuit. We know that: it contains all vertices in the original graph, it is connected, the number of edges in the subgraph is the same as in the vertices, and every vertex in the subgraph has degree 2. Any graph that doesn't possess such a subgraph is, for sure, not Hamiltonian.

Theorem 7 If a graph G is Hamiltonian, then G has a subgraph H with the following properties:

- H contains every vertex of G .
- H is connected.
- H has the same number of edges as vertices.
- Every vertex of H has degree 2.

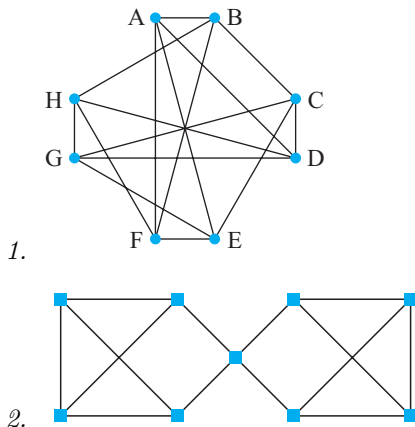
Given below are further conditions that are sufficient to show that a graph is Hamiltonian.

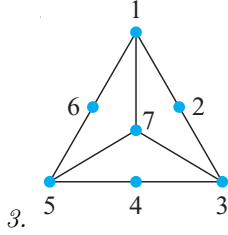
Theorem 8 (Ore's Theorem) If G is a simple graph with $n \geq 3$ vertices such that $\deg(u) + \deg(v) \geq n$ for every pair of nonadjacent vertices, then G has a Hamiltonian cycle.

Theorem 9 (Dirac's Theorem) Hamiltonian Existence from Minimum Degree If G is a simple graph with n vertices and $n \geq 3$ such that the degree of every vertex in G is at least $\frac{n}{2}$, then G is Hamiltonian.

Proof. Since every vertex has degree at least $\frac{n}{2}$, then any two vertices $u, v \in V$ such that u and v are non-adjacent will have $\deg(u) + \deg(v) \geq \frac{n}{2} + \frac{n}{2} = n$. Thus, G is Hamiltonian by Theorem 8.

Example 11 Consider the following graphs. Which of these graphs are Hamiltonian?





Note that graph in first item follows the conditions imposed in Theorem 9. Therefore, one is Hamiltonian. Notice that graph in second item is Hamiltonian although the conditions given in Theorems 8 and 9 are not satisfied. The graph in the third item is not Hamiltonian but contains a Hamilton path.

4 Matrix Representation of Graphs

Here, we learn about another graph representation. This is a commonly used means of describing a graph and provides a way for graphs to be represented inside a computer.

Definition 18 A **matrix** is a two-dimensional sequence of elements. An $m \times n$ matrix A over a set S is a rectangular array of elements of S arranged into rows and columns:

$$A = \begin{pmatrix} a_{11} & \cdots & a_{1j} & \cdots & a_{1n} \\ \vdots & & & & \vdots \\ a_{i1} & & a_{ij} & & a_{in} \\ \vdots & & & \ddots & \vdots \\ a_{m1} & \cdots & a_{mj} & \cdots & a_{mn} \end{pmatrix}$$

A is usually written as $A = (a_{ij})$, $1 \leq i \leq m$, $1 \leq j \leq n$, each $a_{ij} \in S$.

Example 12

$$M = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{pmatrix}$$

1. What is the element at the 2nd row, 4th column?

8

2. What is the 1st column of M ?

$$\begin{pmatrix} 1 \\ 5 \\ 9 \\ 13 \end{pmatrix}$$

3. What is the 3rd row of M ?

$$(9 \ 10 \ 11 \ 12)$$

4. What is the main diagonal of M ?

1, 6, 11, 16

We will explore two types of matrix representation for graphs. One is based on the adjacency of matrices while the other is based on the incidence of vertices and edges.

Definition 19 **Adjacency Matrix**

- An **adjacency matrix** of a simple graph $G = (V = \{v_1, \dots, v_n\}, E)$ is an $n \times n$ matrix, $A = (a_{ij})$, defined as follows:

$$a_{ij} = \begin{cases} 1 & \text{if there is an edge from vertex } v_i \text{ to vertex } v_j. \\ 0 & \text{otherwise} \end{cases}$$

- An adjacency matrix of a pseudograph $G = (V = \{v_1, \dots, v_n\}, E)$, and edge-endpoint function ϵ , is an $n \times n$ matrix, $A = (a_{ij})$, defined as follows:

$$a_{ij} = \text{number of edges from vertex } v_i \text{ to vertex } v_j.$$

Example 13 Given a bipartite graph $G_B = (V_1 \cup V_2, E)$, where

- $V_1 = \{a, b, c, d\}$
- $V_2 = \{e, f, g\}$
- $E = \{\{a, e\}, \{a, f\}, \{a, g\}, \{b, f\}, \{b, g\}, \{c, e\}, \{d, g\}\}$

Construct the adjacency matrix of G_B with respect to the following ordering of the vertices : a, b, c, d, e, f, g .

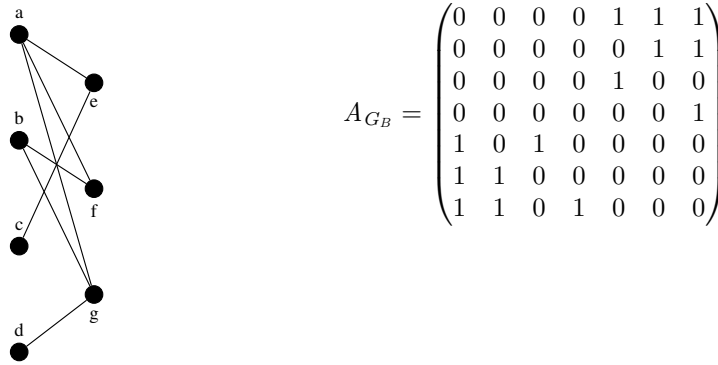


Figure 15: The drawing and matrix representation of G_B in Example 13.

Note that from Definition 19 and based on the mentioned ordering of vertices, we consider $v_1 = a$, $v_2 = b$, $v_3 = c$, $v_4 = d$, $v_5 = e$, $v_6 = f$, and $v_7 = g$. Thus, the value 1 on the 5th column of the 1st row indicates that the edge (a, e) is present in the graph. The value 0 on the 1st row, 2nd column indicates that edge (a, b) is not present.

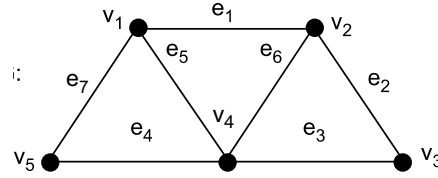
Test Your Understanding

Can a matrix representation of simple graph have main diagonal entries that are non-zero? Why? Why not?

Definition 20 (Incidence Matrix) Suppose that G is an undirected graph with m vertices and n edges, then the incidence matrix of G is an $m \times n$ matrix $B = [b_{ij}]$ where

$$b_{ij} = \begin{cases} 1, & \text{if edge } e_j \text{ is incident on vertex } v_i \\ 0, & \text{otherwise} \end{cases}$$

Example 14 Construct the incidence matrix for the graph below:



Note that the rows will correspond to edges, i.e. e_i will be represented in the i^{th} row. The columns will correspond to the vertices, i.e. v_j will be represented in the j^{th} column.

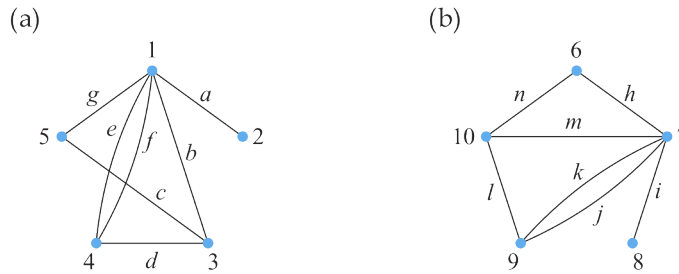
$$\begin{pmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \end{pmatrix}$$

Something to Think About

What will be the matrix representations for directed graphs?

5 Graph Isomorphism

Consider the graphs given below. It looks like these are two different graphs. For example,



we can consider that the names of their vertices and edges are different. However, graph in (a) is very similar to graph in (b). Observe that we can match the following vertices: 1 with 7, 2 with 8, 3 with 10, 4 with 9 and 5 with 6. We can then obtain a similar matching with the edges so that we can obtain one graph from the other. Realizing that two graphs are similar means that there are important properties that are the same on both graphs. Thus, it is important that we can distinguish similar graphs. Thus, we have the next definition:

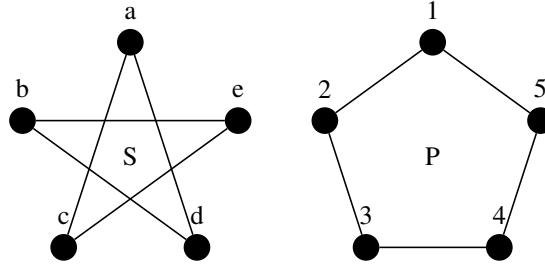
Definition 21 (Graph Isomorphism)

Let $G = (V_G, E_G)$, and $H = (V_H, E_H)$ be graphs.

- A **graph isomorphism** from G to H are bijective functions $p : V_G \rightarrow V_H$, and $q : E_G \rightarrow E_H$ that preserve the edge-endpoint function of G and H . Let $e \in E_G$, and $u, v \in V_G$, and ϵ_G , and ϵ_H be the edge-endpoint functions of G and H respectively. Then,

$$e \mapsto \{u, v\} \in \epsilon_G \text{ iff } q(e) \mapsto \{p(u), p(v)\} \in \epsilon_H$$

- G is **isomorphic** to H , denoted as $G \cong H$, if and only if, there is a graph isomorphism from G to H .

Figure 16: Graph S and P .

Example 15 Consider the two graphs S and P :
Give a graph isomorphism from S to P .

Answer: Let $f : V_S \rightarrow V_P$ be defined as follows: $f = \{a \rightarrow 1, c \rightarrow 2, e \rightarrow 3, b \rightarrow 4, d \rightarrow 5\}$.
Verify that the following holds:

- $\{a, c\} \in E_S$ iff $\{1, 2\} \in E_P$.
- $\{c, e\} \in E_S$ iff $\{2, 3\} \in E_P$.
- $\{e, b\} \in E_S$ iff $\{3, 4\} \in E_P$.
- $\{b, d\} \in E_S$ iff $\{4, 5\} \in E_P$.
- $\{d, a\} \in E_S$ iff $\{5, 1\} \in E_P$.

Since a graph isomorphism is realized for graphs S and P , we say that these graphs are isomorphic.

Theorem 10 Let S be a set of graphs and let $\cong$ be the relation of graph isomorphism on S . Then $\cong$ is an equivalence relation on S .

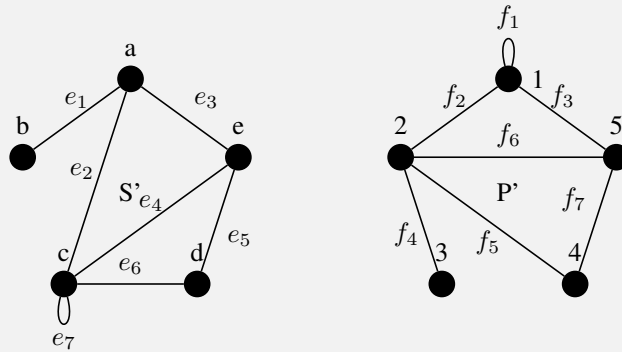
Proof: In order to prove this, we need to show that $\cong$ is reflexive, symmetric and transitive.

- Reflexivity: Let G be a graph in S . To show that $G \cong G$, we can use an identity function both for the vertex set and edge set of G .
- Symmetry: Let G and H be graphs in S . We need to show that if $G \cong H$, then $H \cong G$. Suppose $G \cong H$, then there exists bijective functions g and h for the vertex sets and edge sets from G to H that preserve the edge-endpoint functions. Then g^{-1} and h^{-1} are bijective functions for the vertex sets and edge sets from H and G that preserve the edge-endpoint functions.
- Transitivity: Let G_1 , G_2 , and G_3 be graphs in S . We need to show that if $G_1 \cong G_2$ and $G_2 \cong G_3$, then $G_1 \cong G_3$. Let g_1 and h_1 be the vertex and edge correspondences for the first premise and g_2 and h_2 be the vertex and edge correspondences for the second premise. Then, we can set $g_2 \circ g_1$ and $h_2 \circ h_1$ are vertex and edge set correspondences from G_1 to G_3 that preserve the edge-endpoint functions of G_1 and G_3 .

We can thus group isomorphic graphs into isomorphism classes. Most graph properties, also called *graph invariants*, are then the same for those graphs in the same isomorphism classes.

Test Your Understanding:

Consider the two graphs S' and P' , respectively:



Give a graph isomorphism from S' to P' .

Is there an algorithm to determine whether two graphs are isomorphic, i.e. $G \cong H$? Yes, let's say that G and H each have n vertices. We can for example, consider first all possible correspondences for their vertices. How many possible correspondence should we consider? This will be $(n!)$. In later subjects, you will learn that such exhaustive method, although correct, is not efficient.

Unfortunately, there is no known efficient general method for checking if two graphs are isomorphic. However, there are simple ways to determine if two graphs are non-isomorphic. For example, a pair of graphs that do not have the same number of vertices cannot be isomorphic.

Let us now talk about properties that are preserved in isomorphic graphs.

Definition 22 (Graph Invariants)

A property P is called an **invariant for graph isomorphism** if, and only if, given any graph G and H , if G has property P and $G \cong H$, then H has property P .

Remark: Given an invariant property P , if G has property P and H does not have property P , then $G \not\cong H$.

Theorem 11 Each of the following properties is an invariant for graph isomorphism, where n , m , and k are all nonnegative integers:

1. has n vertices;
2. has m edges;
3. has a vertex of degree k ;
4. has m vertices of degree k ;
5. has a circuit of length k ;
6. has a simple circuit of length k ;
7. has m simple circuits of length k ;
8. is connected;
9. is Eulerian;
10. is Hamiltonian.

Example 16 Consider the two graphs G_1 and G_2 in Figure 17. The two graphs are not isomorphic. Find a property in which one graph has and the other does not.

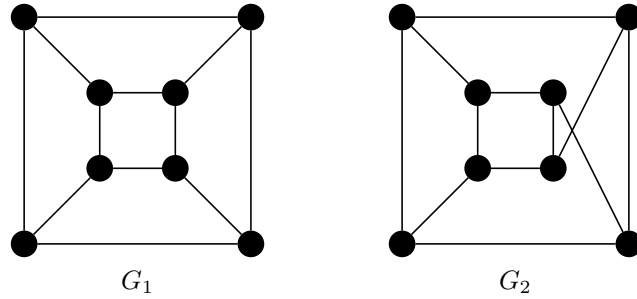


Figure 17: Graph G_1 and G_2 .

One property violated is as follows: Observe that in G_1 , there are 6 simple circuits of degree 4. However, in G_2 , there are only 4 simple circuits of degree 4.

Example 17 Consider the two graphs G_3 and G_4 in Figure 18. Are the graphs isomorphic? If yes, give the isomorphism. If no, why?



Figure 18: Graph G_3 and G_4 .

Observe that there are two circuits of length 6 in G_3 . However, in G_4 , there is only one such circuit. Also, there is a circuit of length 8 in G_4 . This doesn't exist in G_3 .

6 Graph Operations

Definition 23 Consider a graph $G = (V_G, E_G)$

- Let $v \in V_G$ be a vertex. The operation $G - \{v\}$ is a graph defined as

$$G - \{v\} = (V_G - \{v\}, E_G - \{\{v', v\} \mid \forall v' \in V_G\}).$$

- A vertex v in G is said to be a **cut vertex**, if $G - \{v\}$ has at least one more component than G .
- Let $e \mapsto \{u, v\}$ be an edge, and $u, v \in V$. Then $G - \{e\}$ is defined as

$$G - \{e\} = (V_G, E_G - \{e\})$$

- An edge e in G is said to be a **cut edge**, if $G - \{e\}$ has at least one more component than G .
- If G is a simple graph, the **complement** of G , denoted as G^c , is the graph $G^c = (V_G, E_{G^c})$ where $E_{G^c} = \{e \mid e \in \mathbb{P}_2(V_G), e \notin E_G\}$. ($\mathbb{P}_2(V)$ is the set of subsets of V where each subset has exactly 2 elements.)

The following theorem shows that the complement graphs of isomorphic graphs are also isomorphic.

Theorem 12 Let G and H be simple graphs. If $G \cong H$, then $G^c \cong H^c$.

Proof. Since $G \cong H$, there exists bijective functions g and h for the vertex sets and edge sets from G to H that preserves the edge-endpoint functions. We can then use the same bijective function g for the vertex correspondence from G^c to H^c . For the bijective function h' for edge sets from G^c to H^c , given an edge e with endpoints u and v , $h'(e)$ will be the edge with end points $g(u)$ and $g(v)$. Since $G \cong H$, vertices u and v are adjacent in G if and only if vertices $g(u)$ and $g(h)$ are adjacent in H . This also means, vertices u and v are not adjacent in G if and only if $g(u)$ and $g(h)$ are not adjacent in H . Then, u and v are adjacent in G^c if and only if $g(u)$ and $g(h)$ are adjacent in H^c . Thus, the edge-endpoint function is also preserved in functions g and h' .

Example 18 Consider the graph G in Figure 19 and the resulting graphs from the defined graph operations.

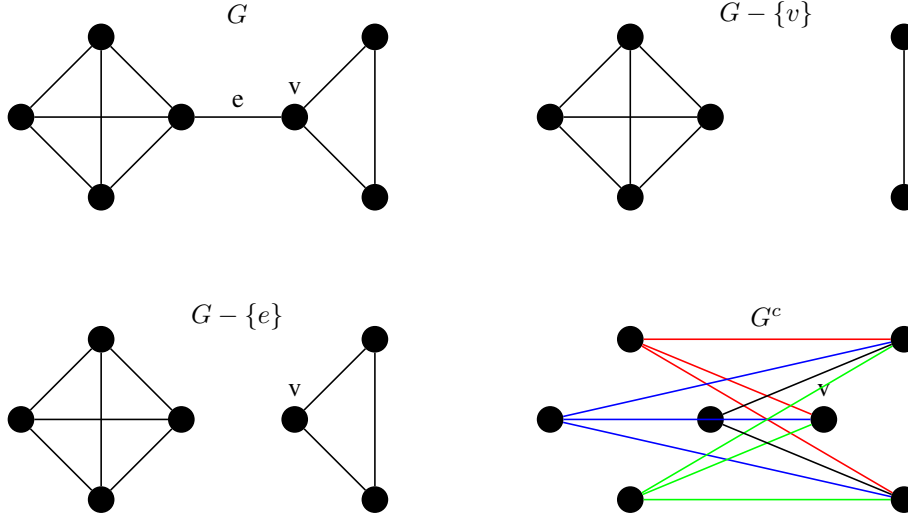


Figure 19: Graph operation on G .

Observe that $G - \{v\}$ is a cut vertex and $G - \{e\}$ is a cut edge. While graph G is only composed of one component, both $G - \{v\}$ and $G - \{e\}$ have two components in their resulting graphs.

7 Graph Connectivity

Suppose we have a graph model of a computer network. We know that any two computers in the network can communicate if the graph is connected. However, we also want to determine how reliable is the network, i.e. suppose several communication link fails, will all the computers still be able to communicate with each other? To answer such questions, we will need the following concepts:

Definition 24 Let $G = (V, E)$ be a connected graph.

- Let $W \subseteq V$, and $F \subseteq E$. Then $G - (W \cup F)$ is a graph, G' , defined as,

$$G' = (V - W, E - (F \cup \{e \in E \mid e \mapsto \{u, v\}, v \in W \vee u \in W\}))$$

- A disconnecting set of vertices for G is a set of vertices, D , such that $G - D$ is disconnected.

- The vertex connectivity of G , denoted as $\kappa(G)$, is the minimum number of vertices whose removal results in either a disconnected graph, or a trivial graph.
- A κ -set for G is a set of $\kappa(G)$ vertices whose removal results in either a disconnected graph, or a trivial graph.
- A disconnecting set of edges for G is a set F such that $G - F$ is disconnected.
- The edge connectivity of G , denoted as $\lambda(G)$, is the minimum number of edges whose removal results in either a disconnected graph, or a trivial graph.
- A λ -set is a disconnecting set of $\lambda(G)$ edges.
- Note that if G has only one vertex, then $\lambda(G) = |E|$.

Example 19

Determine the κ connectivity and λ connectivity of the following graphs:

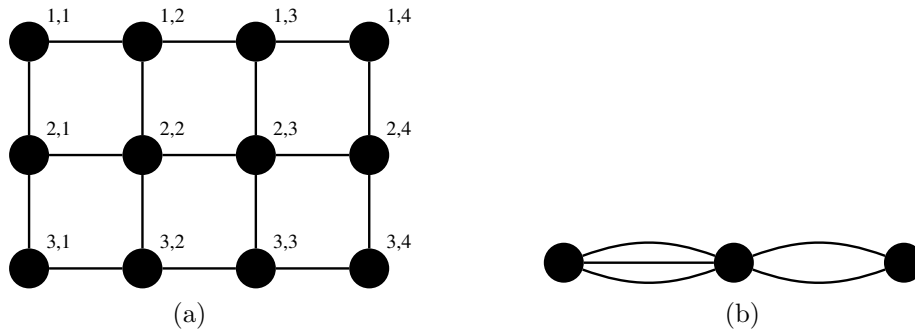


Figure 20: Drawing in (a) represents a 3×4 mesh graph while drawing in (b) represents a multigraph.

1. A path graph, P_n , of $n \geq 2$ vertices.
2. A cycle graph, C_n , of $n \geq 3$ vertices.
3. A complete graph, K_n , of $n \geq 2$ vertices.
4. A p -regular graph, with $p \geq 2$.
5. A complete bi-partite graph of $m \times n$ vertices, $K_{m,n}$, with $m, n \geq 1$.
6. The 3×4 mesh graph in Figure 20.(a).
7. The multigraph in Figure 20.(b).

G	$\kappa(G)$	$\lambda(G)$
P_n	1	1
C_n	2	2
K_n	$n - 1$	$n - 1$
p -regular	p	p
$K_{m,n}$	$\min\{m, n\}$	$\min\{m, n\}$
Fig. 20.(a)	2	2
Fig. 20.(b)	1	2

One way to make a graph disconnected is to isolate one vertex. This can be done by removing all of its neighbors or removing all its incident edges. Thus, we have the following theorem.

Theorem 13 Minimum Degree-Connectivity

Let G be a graph and $\delta(G)$ denote the degree of the vertex in G with the least number of edges incident to it. Then,

$$\begin{aligned}\kappa(G) &\leq \delta(G) \\ \lambda(G) &\leq \delta(G)\end{aligned}$$

Given a graph, removing one vertex also involves removing all the edges incident to the vertex. Thus, suppose we have a set of edges that is a λ -set for a graph G . We can choose one endpoint for each edge in the λ -set to form a set of vertices that also disconnects graph G . Thus, $\kappa(G) \leq \delta(G)$. Therefore, we have the following theorem.

Theorem 14 Let G be a graph. Then, $\kappa(G) \leq \lambda(G) \leq \delta(G)$.

Theorem 15 Hamiltonian Connectivity

Let G be any graph. If G has a Hamilton circuit, then $\kappa(G) \geq 2$.

Proof. There is a subgraph of G that is isomorphic to a cyclic graph C_n covering all vertices of G . Thus, $\kappa(G) \leq \kappa(C_n)$. Two vertices are needed to disconnect C_n , i.e. $\kappa(C_n) = 2$. Thus, $\kappa(G) \leq 2$.

We can also use the contrapositive as a criteria to conclude when a graph has no Hamilton circuit, i.e. if $\kappa(G) \leq 1$, then G is not Hamiltonian.

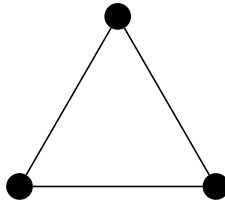
8 Graph Planarity

In application to circuits, we can model a circuit board as a graph. The edge crossings can then be thought of as wire connections. However, designing a circuit board is not trivial because the number of connections increase with the number of components. The question then is: can we create an optimal layout given the number of components and connections? An ideal layout is one in which there are no wire crossings. In graph theory, this is represented by so-called planar graphs.

Definition 25 A graph is called **planar** if it can be drawn in the plane without any edges crossing (where a crossing of edges is the intersection of the lines or arcs representing them at a point other than their common endpoint). Such a drawing is called a **planar representation** (also, a **planar embedding**) of the graph.

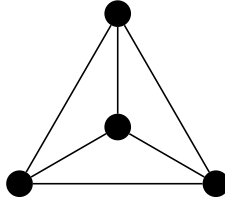
Example 20 Find a planar embedding, if any, for the following graphs.

1. K_3 .



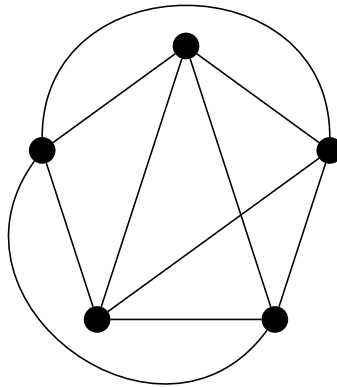
Hence, K_3 is planar.

2. K_4 .



Hence, K_4 is planar.

3. K_5 . Actually, K_5 does not have a planar embedding. The best is a drawing with at most one crossing.



So is K_5 non-planar?

Note that we can show that a graph is planar by showing a planar representation. However, it is harder to show that a graph is non-planar. The existence of a drawing with edge crossing doesn't necessarily imply that a graph is non-planar. We need to have stronger tools that can be used to prove a negative result.

8.1 Dual Graph

Definition 26 A planar embedding of a graph $G = (V_G, E_G)$ splits the plane into **regions**, including an unbounded region.

- Let R_G denote the set of regions of G .
- The dual graph of a planar embedding of G , $D(G)$, is the graph with vertex set corresponding to R_G , and edges connect two vertices if the regions represented by these vertices share an edge as their border.

Given in Figure 21 is an example of a planar graph with its dual graph. Observe that each dotted edge crosses exactly one solid edge and vice versa. Also, note that for any graph G with a planar embedding, $D(G)$ is always planar and $D(D(G)) \cong G$.

8.2 Planarity of Graphs

Theorem 16 (Euler's Formula)

Given any planar embedding of a connected graph $G = (V, E)$ with region set R ,

$$|V| - |E| + |R| = 2$$

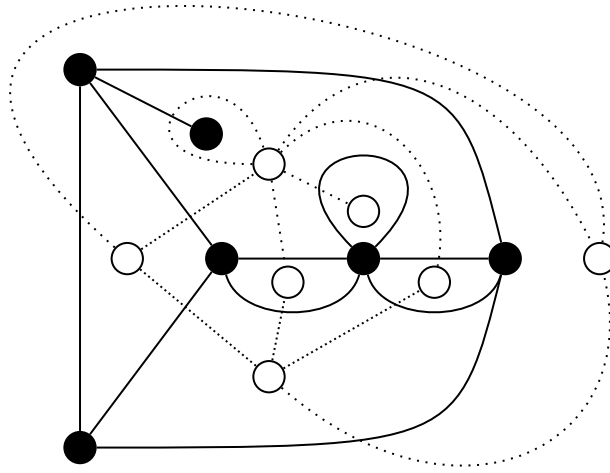


Figure 21: A graph and its dual graph.

Proof: Let $G = (V, E)$ be a graph with region set R . Let $n = |V|$. The proof is by induction on the number of vertices of G . Since V should be non-empty, the base case is $n = 1$.

Base Case ($n = 1$): If $n = 1$, then all edges must be loops. Hence $|R| = |E| + 1$. Substituting the values to the formula,

$$\begin{aligned} 1 - |E| + |E| + 1 &= 2 \\ 2 &= 2 \end{aligned}$$

Inductive Step: For a $n \geq 1$, let's assume that Euler's Formula holds for graphs of n vertices. Suppose G is a graph with $n + 1$ vertices and with a planar embedding. We need to show that Euler's Formula still holds for G .

Since G is connected, there exists a non-loop edge $e \mapsto \{u, v\}$. Form a new graph $G' = (V', E')$ from G by removing e and combining u and v into a single vertex v' . The planar embedding of G' has region set R' , where $|R'| = |R|$. Since $|V'| = |V| - 1 = n$, and $|E'| = |E| - 1$, then by the inductive hypothesis,

$$\begin{aligned} |V'| - |E'| + |R'| &= 2 \\ |V| - 1 - (|E| - 1) + |R| &= 2 \\ |V| - |E| + |R| &= 2 \end{aligned}$$

Therefore, Euler's Formula holds for any graph G of n vertices. ■

Corollary 3 Given a connected planar simple graph $G = (V, E)$ with $|V| \geq 3$,

$$|E| \leq 3|V| - 6$$

Proof. The dual graph of G , $D(G)$, has a vertex set R and edge set E' where each edge $e' \in E'$ corresponds to an $e \in E$ so that $|E'| = |E|$. From Theorem 1, $\sum_{r \in R} \deg(r) = 2|E|$.

Since G is simple, it has no regions bounded by just one or two edges. (Each region is bounded of three or more edges). Thus, for each region r , $\deg(r) \geq 3$. Thus, $3 \mid R \mid \leq 2 \mid E \mid$.

From Theorem 16,

$$\begin{aligned} 2 \mid V \mid - \mid E \mid + \mid R \mid &\leq \mid V \mid - \mid E \mid + \frac{2}{3} \mid E \mid = \mid V \mid - \frac{1}{3} \mid E \mid \\ 6 &\leq 3 \mid V \mid - \mid E \mid \\ \mid E \mid &\leq 3 \mid V \mid - 6 \end{aligned}$$

■

Corollary 4 *Given a connected planar simple graph $G = (V, E)$ with $\mid V \mid \geq 3$, and G does not contain cycles of length 3, then*

$$\mid E \mid \leq 2 \mid V \mid - 4$$

The proof for Corollary 4 is almost the same as in Corollary 3. However, since there are no simple circuits of length 3, all regions is bounded by at least four edges which means $\deg(r) \geq 4$ and thus, $4 \mid R \mid \leq 2 \mid E \mid$.

Example 21 *Use the two corollaries to prove that K_5 and $K_{3,3}$ are non-planar.*

Corollary 5 *If G is a connected planar simple graph, then G has a vertex of degree not exceeding five.*

Proof. If G has one or two vertices, the result is true. If G has at least three vertices: From Corollary 3, $\mid E \mid \leq 3 \mid V \mid - 6$ so $2 \mid E \mid \leq 6 \mid V \mid - 12$. If a graph G has vertices with degree of at least six, From the Theorem 1, since $2 \mid E \mid = \sum_{v \in V} \deg(v)$, $2 \mid E \mid \geq 6 \mid V \mid$. We will then have, $6 \mid V \mid \leq 2 \mid E \mid \leq 6 \mid V \mid - 12$, which is not possible.

■

Example 22 K_n , $n \geq 7$, are non-planar.

Definition 27 *Given a graph $G = (V, E)$, and edge $e \mapsto \{u, v\}$ of G , a new graph G' can be formed from G by removing e from E , and adding $e' \mapsto \{u, w\}$ and $e'' \mapsto \{w, v\}$. Making two edges from one by adding a vertex between the two endpoints is referred to as an **edge subdivision**.*

*A graph G' is a **subdivision** of G if G' is obtained through a sequence of zero or more edge subdivisions of G .*

Two graphs G' and G'' are homeomorphic if there exists a graph G such that G' and G'' are subdivisions of G .

Theorem 17 (Kuratowski's Theorem) *A graph G is non-planar if and only if G contains a subgraph that is homeomorphic to either K_5 or $K_{3,3}$.*

We can easily argue that any graph that contains a subdivision of either $K_{3,3}$ or K_5 is nonplanar. However, the proof of the other direction is outside the scope of this lesson.

Example 23 *The graph in Figure 23 is not planar since it contains a subgraph homeomorphic to K_5 .*

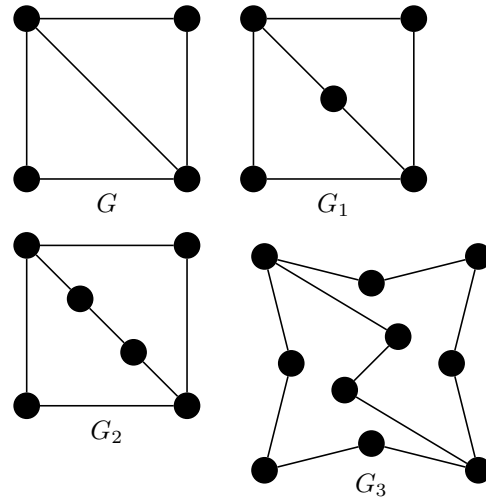


Figure 22: A graph and some of its homeomorphism.

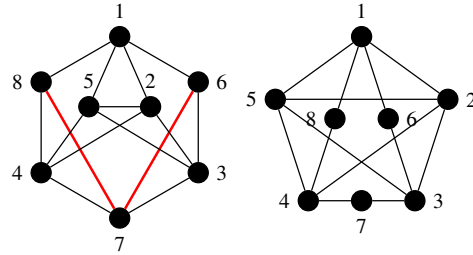


Figure 23: The graph in the left is an example of a non-planar graph. Notice that if we remove the two bold edges, the resulting subgraph shown in the right is a subdivision of K_5 . Figure is adapted from (Ferland, 2009)

9 Graph Coloring

Graphs can also be used to model maps. When dealing with maps, we treat each region as a separate vertex and connect two vertices whenever they share the same boundary. One interesting problem in maps is the use of colors to distinguish the boundaries of each region. It is common to use two different colors for adjacent regions and one way to ensure that no two adjacent region uses the same color is to simply use a different color each region different. However, this will be inefficient and in the case of many regions, it is possible that some colors are hardly distinguishable. Thus, it is desirable to only use a small number of colors. This leads to the question, what can be the minimum number of colors that we can use to color a graph. This is represented by the concept of chromatic number in graph theory.

Definition 28 Let $G = (V, E)$ be a graph.

- A **coloring** is an assignment of colors to the vertices of G in such a way that no two adjacent vertices have the same color.
- A **color class** for a coloring is a set of all the vertices of one color. The vertices are partitioned by the color classes.
- A **k -coloring** of G means that k colors are used for coloring G .

- G is **k -colorable** if there exists a coloring of G that uses at most k colors.
- $\chi(G)$, denotes the **chromatic number** of G . It is the minimum number of colors to color G .

Note that no coloring exists for graphs with loops. Note that parallel edges do not affect coloring.

Definition 29 Let G be a graph.

- A **clique** in G is a subgraph that is complete.
- The **clique number** of G , denoted as $\omega(G)$, is the maximum number of vertices in a clique of G .

Note that a complete graph K_n will need n colors since all vertices are adjacent to everyone. Thus, there is a relation between the maximum number of vertices in a clique of a graph G and the minimum number of colors to color G . Specifically, we have the following theorem.

Theorem 18 Let G be any graph without loops. Then $\chi(G) \geq \omega(G)$.

Example 24 A graph coloring of a graph with a 5-clique is given in Figure 24.

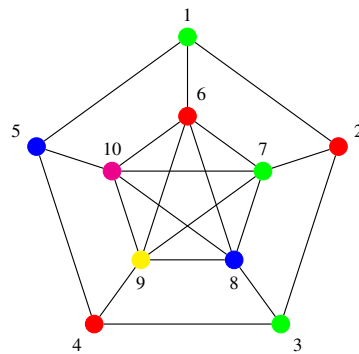


Figure 24: A coloring of a graph with a 5-clique.

Theorem 19 (Four Coloring Theorem) Any planar graph G has $\chi(G) \leq 4$.

The theorem is notable in the history of computer science as it is the first computer-aided proof.

Coloring Maps

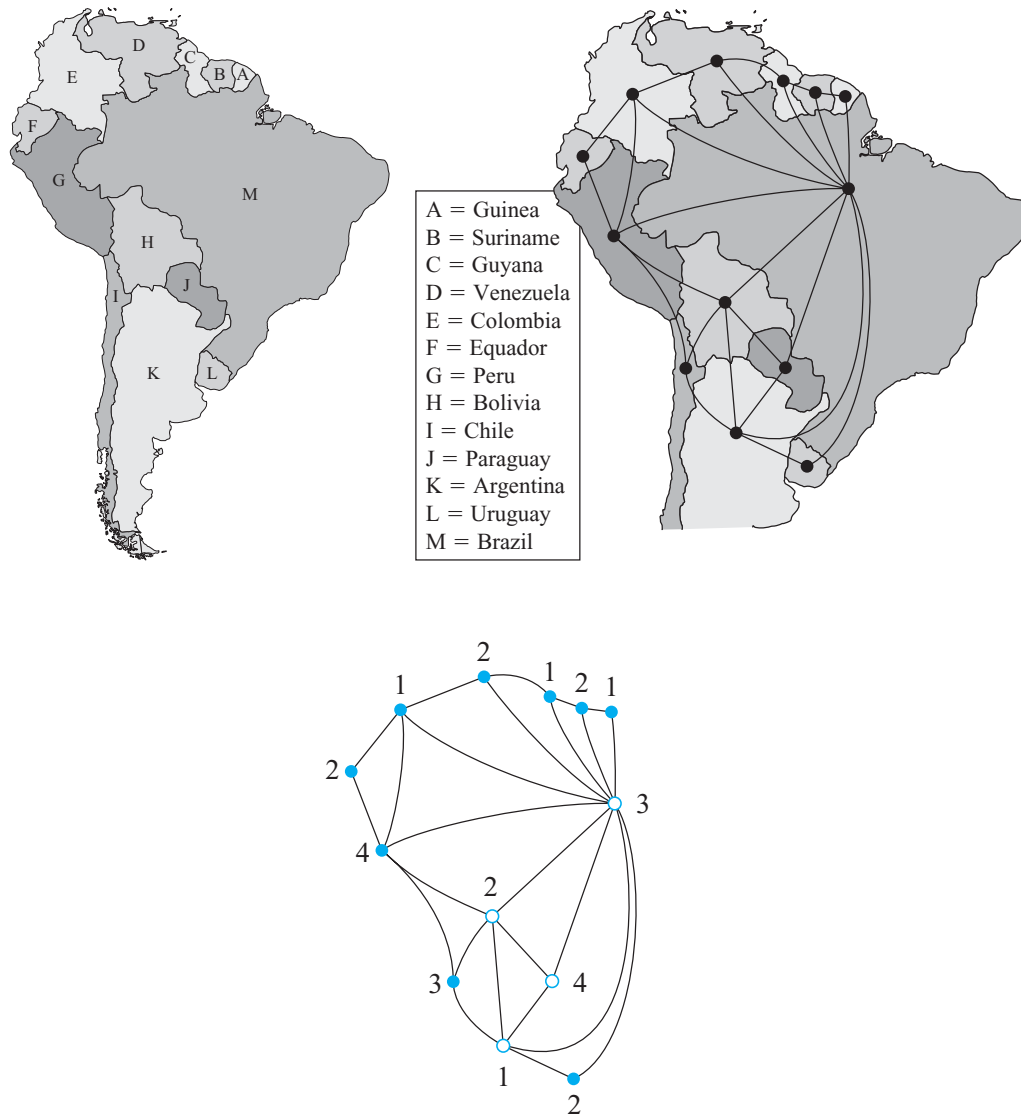


Figure 25: A coloring of the map of South America

10 Trees

“I think that I shall never see, a poem lovely as a binary tree.” - Joyce Kilmer

Definition 30 A graph is said to be **circuit-free** if and only if it has no circuits. A graph is called a **tree** if, and only if, it is circuit-free and connected. A **trivial tree** is a graph that consists of a single vertex. A graph is called a **forest** if, and only if, it is circuit-free and disconnected. Each of the component of a forest is a tree.

Definition 31 Let T be a tree. If T has only one or two vertices, then each is called a **terminal vertex**. If T has at least three vertices, then vertices of degree 1 in T are terminal vertices (or **leaves**), and vertices of degree greater than 1 are called **internal (branch) vertices**.

Lemma 1 *Any tree that has more than one vertex has at least one vertex of degree 1.*

Proof. Let T be a tree. Let P be the longest simple path in T . The start and end vertices of P must be distinct and must have degree 1. Otherwise, P can be made longer. ■

Theorem 20 *For any positive integer n , any tree with n vertices has exactly $n - 1$ edges.*

Proof. We can use a proof by mathematical induction.

- *Base Case:* Verify that the result is true when $n = 1$ and $n = 2$.
- *Inductive Step:* Assume that the result is true for n . We need to show that the result is also true for $n + 1$.

Let T be a tree with $n + 1$ vertices. We know that T has at least one vertex of degree 1 from Lemma 1. Let's name this vertex v and let e be the edge incident with v .

We now define a subgraph T' from T obtained by removing v and e from T . We thus know that T' has $(n + 1) - 1$ vertices. We also know that T' is circuit-free and T' is still connected. Therefore, T' is a tree. Following our assumption, T' will have $n - 1$ edges. Thus, T has $(n - 1) + 1$ edges. ■

Theorem 21 *For any positive integer n , if G is a connected graph with n vertices and $n - 1$ edges, then G is a tree.*

Definition 32 *A **rooted tree** is a tree in which there is one vertex that is distinguished as the **root**. The **level** of a vertex is length of the unique path between it and the root. The **height** of the rooted tree is the maximum level of any vertex of the tree. For any internal vertex v or the root of a rooted tree, the **children** of v are the vertices that are adjacent to v and are one level away from the root than v . If w is a child of v , then v is the **parent** of w . Two distinct vertices, u and v that are both children of the same parent are called **siblings**. Given two distinct vertices v and w , if v lies on the unique path between w and the root, then v is an **ancestor** of w and w is a **descendant** of v .*

Definition 33 *A **binary tree** is a rooted tree in which every parent has at most two children. Each child in a binary tree is designated either as **left child** or **right child** (but not both), and every parent has at most one left child and one right child. A **full binary tree** is a binary tree in which each parent has exactly two children.*

*Given any parent v in a binary tree T , if v has a left child, then the **left (right) subtree** of v is the binary tree whose root is the left (right) child of v , whose vertices consists of the left (right) child of v and all its descendants, and whose edges consists of all edges of T that connect the vertices of the left (right) subtree.*

Theorem 22 *If k is a positive integer and T is a full binary tree with k internal vertices, then T has a total of $2k + 1$ vertices and has $k + 1$ terminal vertices.*

Proof. Every vertex, except the root, is the child of an internal vertex. Each of the k internal vertices has 2 children, thus, there are $2k$ vertices except the root. Then, including the root, there are $2k + 1$ vertices. Since the number of vertices is the number of internal vertices plus the number of terminal vertices,

$$2k + 1 = k + m$$

where m is the total number of terminal vertices. The value of m is then equal to $k + 1$. ■

Theorem 23 For all integers $h \geq 0$, if T is any binary tree of height h and t terminal vertices, then

$$t \leq 2^h$$

Equivalently, $\log_2 t \leq h$.

Proof. We will use a proof by strong induction on h .

Verify that the result is true in the base case, i.e. when $h = 0$.

For the inductive step, let us assume that result is true for all $h \geq 0$. We need to show that the result is also true for $h + 1$. Let T be a binary tree of height $h + 1$. Let v_0 be the root of T . If we remove v_0 , then there remain at most two rooted trees, where each rooted tree has a child of v_0 as its root. Also, the rooted tree is a binary tree of height at most h . The inductive hypothesis can then be applied to the rooted trees obtained. Note that the terminal vertices of T is equal to the total number of leaves in the rooted trees. Since there can be at most two rooted trees of height at most h , the total number of leaves in T is at most $2^h + 2^h = 2^{h+1}$.

From $t \leq 2^{h+1}$, we take the $\log_2$ of both sides to obtain, $\log_2 t \leq h + 1$.

■

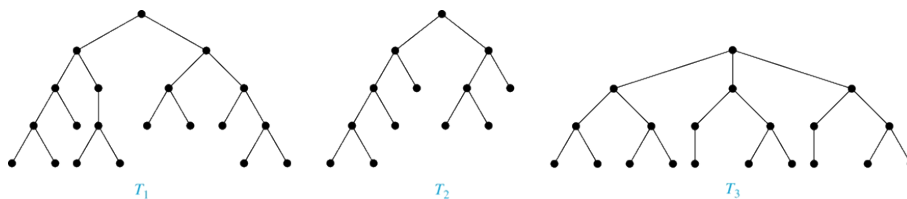
Definition 34 A rooted tree is called an m -ary tree if every internal vertex has no more than m children. The tree is called a full m -ary tree if every internal vertex has exactly m children.

Test Your Understanding

- How many vertices does a full m -ary tree with k internal vertices have?
- What is an upper bound for the number of leaves in an m -ary tree of height h ?

Definition 35 A rooted m -ary tree of height h is balanced if all leaves are at levels h or $h - 1$.

Example 25 Which of the rooted trees shown below is balanced?



11 Some Graph-related Hard Problems

11.1 Clique Problem

The computational problem of finding cliques for an input graph. Some formulation of the clique problem include the **Maximum Clique** optimization problem, which is to determine $\omega(G)$, i.e. finding the largest clique in graph G . Another well-known formulation is the **Decision Clique Problem**, the problem of testing whether a graph has a clique of a given size.

An exhaustive (i.e. brute-force) algorithm for solving the decision problem version given a clique size of k would be to check all $\binom{n}{k}$ vertex subsets of the graph and check if the

subgraph containing these k vertices form a clique. For the maximum clique, the algorithm check for all

$$\sum_{k=1}^n \binom{n}{k}$$

vertex subsets and determine which of these vertex subsets form the largest clique (if there is one at all).

11.2 Vertex Cover Problem

A **vertex cover** of a graph is a set of vertices such that each edge of the graph is incident to at least one vertex in the vertex cover set. The Vertex Cover problem is a decision problem to determine whether a graph has a vertex cover with cardinality less than or equal to a given size. The optimization version, **Minimum Vertex Cover**, is to find the minimum number of vertices needed to cover all edges.

Similar to the Clique Problem, the best known algorithm for the Vertex Cover problem is an exhaustive search.

11.3 Independent Set Problem

An **independent set** of a graph is a set of vertices wherein no vertices are adjacent to each other, i.e. vertices in the independent set do not have edges in between them. The decision problem would be to determine if a graph has an independent set of cardinality greater than or equal to a given size. The optimization problem would be **Maximum Independent Set**, the problem of finding the largest possible size for an independent set for an input graph G . This size is called the independence number of G , denoted as $\alpha(G)$.

Again, similar to the two previous problem, the best known algorithm for the Independent Set Problem is an exhaustive search for any arbitrary input graph.

12 Exercises

Think about these:

1. Prove that every simple circuit in a complete bipartite graph has even length.
2. For what values of m, n does the complete bipartite graph $K_{m,n}$ have an Euler circuit? a Hamilton circuit? Why?
3. Suppose that a graph G with n vertices is acyclic (it contains no subgraph that has a simple circuit) but not necessarily connected. Prove that if G has k connected components then it has $n - k$ edges.
4. Given the simple graph H with vertex set $\{a, b, c, d, e\}$ and nine edges consisting of all possible edges except edge $\{c, e\}$. Are all simple graphs with five vertices and nine edges isomorphic to H ? Why?